
Advanced signal processing

Exam 1

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Due date: 4/10/2026

1 Problem 1

Consider a causal system given by the following system function

$$H(z) = 1 + 3.5z^{-1} + 1.5z^{-2}$$

And assume that the causal system is used as a model for a communication channel that will distort signals passing through it.

1.1 1.

Design an equalizer that can mitigate the distortion The system function is being described as causal, meaning that it cannot depend on future inputs, only on previous inputs $x[n], x[n-1], x[n-2], \dots$. In non-minimum-phase systems, the distortion from systems $H(z)$ can be *undone* by cascading an equalizer, or filter, with the system function, such that when multiplied by its inverse, its effects will be undone. $H(z)H_{inv}(z) = 1$. From the given system function, the zeros and poles can be extracted. A stable and causal system, should have poles inside the unit circle, but the zeros does not need to be. To find out what kind of equalizer we should construct, we can firstly analyze our system to find these.

$$H(z) = 1 + 3.5z^{-1} + 1.5z^{-2} \Rightarrow H(z)z^2 = z^2 + 3.5z^1 + 1.5 \Rightarrow H(z) = \frac{z^2 + 3.5z^1 + 1.5}{z^2}$$

rewritten as

$$\frac{(z+3)(z+0.5)}{z^2}$$

yielding zeroes in: $z = -3, z = -0.5$. Here one zero lies within the unit circle, while the other does not. This is not a problem for the original system.

Our equalizer should ideally be an inverse of the original system, such that

$$H_{inv} = \frac{1}{H(z)} = \frac{A(z)}{B(z)}$$

such that

$$H(z) = \frac{B(z)}{A(z)}$$

however because of this, the zeroes from the original becomes the poles, which now resides outside the unit circle making it unstable. Therefore we must decompose this system into a minimum-phase system and an allpass system.

$$H(z) = H_{min}(z)H_{ap}(z)$$

taking the deduced system function from earlier, this can be rewritten into an expression, that can be made into a system function that will be composed of our minimum-phase and all-pass systems

$$H(z) = \frac{(z+3)(z+0.5)}{z^2} \Rightarrow \frac{z+0.5}{z} \frac{z+3}{z} \Rightarrow (1+0.5z^{-1})(1+3z^{-1})$$

In this case, we have a zero $z = -3$ outside the unit circuit, and $z = \frac{1}{a}, |a| < 1 \Rightarrow a = -\frac{1}{3}$ The composition can be written as:

$$H(z) = H_1(z) \underbrace{(1 - az^{-1})}_{\text{minimum-phase factor}} \underbrace{\frac{z^{-1} - a^*}{1 - az^{-1}}}_{\text{All-pass factor}}$$

$\Downarrow$

$$H(z) = 3(1+0.5z^{-1})(1+\frac{1}{3}z^{-1})\frac{z^{-1}+\frac{1}{3}}{1+\frac{1}{3}z^{-1}}$$

Where $H_1(z) = (1 + 0.5z^{-1})$ and the additional factor of 3, added to the front of $H_1(z)$ is the scale factor, which the magnitude of a.

cleaning up the expressiong yields:

$$H(z) = 3(1 + 0.5z^{-1})(z^{-1} + \frac{1}{3})$$

All-pass filters affect phase, but not magnitude, therefore we can construct our filter as

$$H_{eq}(z) = \frac{1}{H_{min}} = \frac{1}{3(1 + 0.5z^{-1})(1 + \frac{1}{3}z^{-1})}$$

As mentioned ealier, there is a relationship between the input system, the minimum phase and the all pass function. Since we have the inverse of the minimum phase, we can multiply the equalizer with the input system and then get the all pass function.

$$\begin{aligned} H(z)H_{eq}(z) &= 3(1 + 0.5z^{-1})(1 + \frac{1}{3}z^{-1}) \frac{z^{-1} + \frac{1}{3}}{1 + \frac{1}{3}z^{-1}} \cdot \frac{1}{3(1 + 0.5z^{-1})(1 + \frac{1}{3}z^{-1})} \\ &\Downarrow \\ &= \frac{3(1 + 0.5z^{-1})(z^{-1} + \frac{1}{3})}{3(1 + 0.5z^{-1})(1 + \frac{1}{3}z^{-1})} \\ &\Downarrow \\ &= \frac{(z^{-1} + \frac{1}{3})}{(1 + \frac{1}{3}z^{-1})} = H_{ap}(z) \end{aligned}$$

And this expression is equal to our allpass system.

The magnitude- and phasereponse is shown below

1.2 2.

Make a time-domain simulation of the system and the equalizer using a test signal of your choice. Explain the choice of test signal and procedures .. analyze the performance of the equalizer

A good test signal for the system, with the equalizer, would be a squarewave signal as it contains its fundamental frequency, as well as, essentially, infinitely many odd harmonics thereof. We'll select a signal with frequency of $f = 2kHz$ and a sampling rate $f_s = 48kHz$

1.3 3.

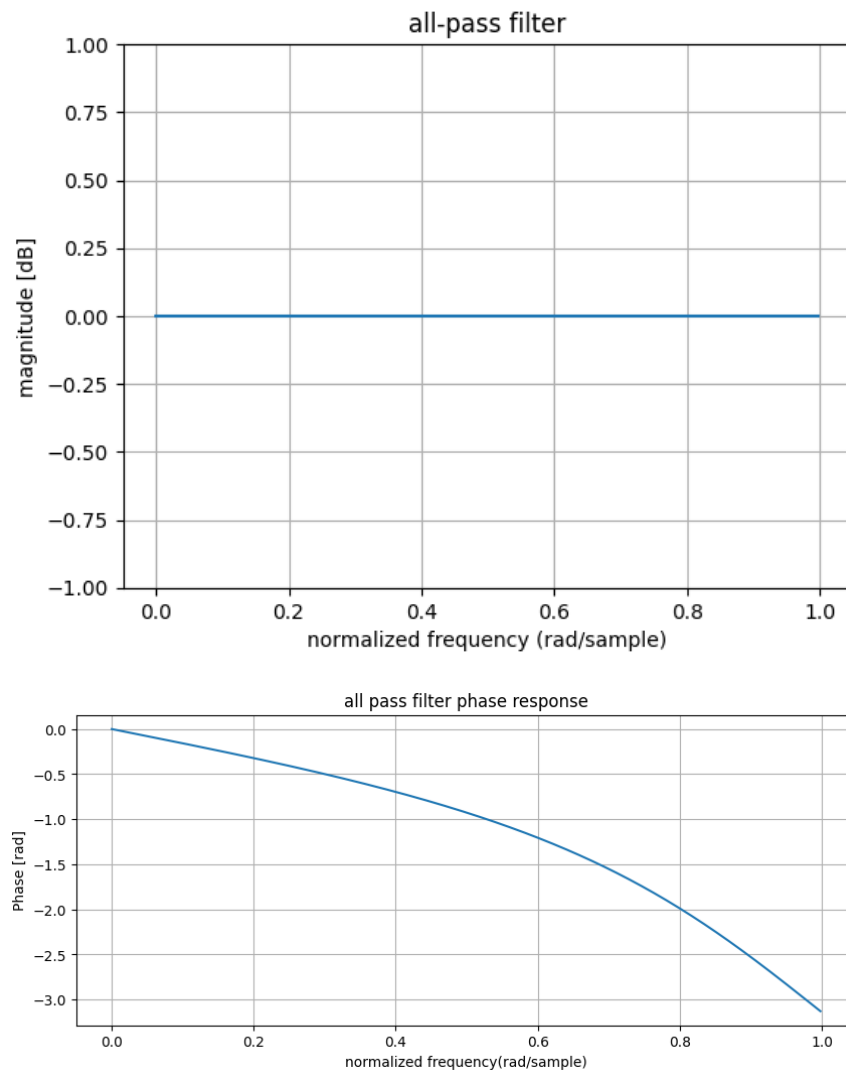


Figure 1: magnitude and phasereponse of the deduced all-pass filter

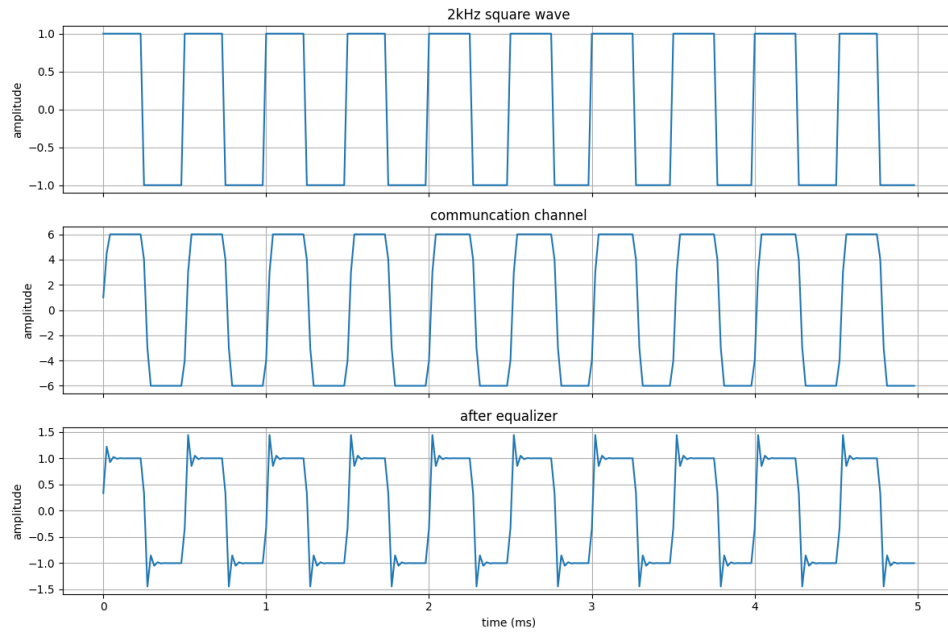


Figure 2: 2kHz testsignal, passed through, first the communcation channel and the through the equalizer. Note the slight overshoot of the signal edges from the rise and fall times.

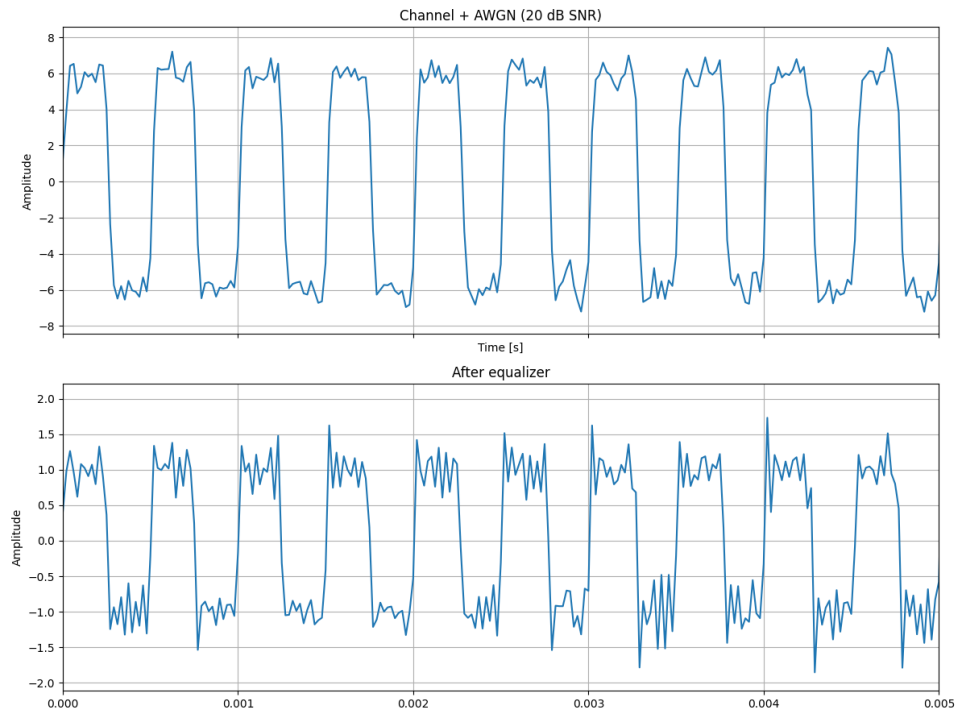


Figure 3: white gaussian noise added after the communcation channel, in this case the noise is 20dB SNR. Before the eq, the signal waveform can easily be made out, however after the eq, some sharpness has been retained in the rise and falling edges, however distortion is still present.